Bottle Detection System Architecture Technical Architecture Documentation

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Technical Architecture Documentation

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Executive Summary

This document presents the technical architecture of an automated Bottle Detection System utilizing the YOLOv8 deep learning model. The system processes input images through a multi-stage pipeline to accurately identify and localize bottles within visual data. This architecture documentation outlines each processing stage, from initial image preprocessing through final visualization output. The solution provides real-time object detection capabilities with high accuracy and confidence-based filtering mechanisms. This system is designed for industrial quality control, inventory management, and automated monitoring applications.

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1. Document Overview

1.1 Purpose

This architecture document provides a comprehensive technical overview of the Bottle Detection System's design and implementation. It serves as a reference for developers, engineers, and stakeholders to understand the system's components, data flow, and processing stages.

1.2 Scope

The document covers the complete detection pipeline from image input to final output visualization. Each architectural component is detailed with its purpose, inputs, outputs, and technical advantages.

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2. System Architecture Diagram

📷 INPUT IMAGE

Raw image data from camera/file system

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🔧 IMAGE PREPROCESSING

Normalization, resizing, augmentation

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🧠 YOLOv8 MODEL

Deep learning object detection

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📦 BOUNDING BOXES

Detected object coordinates

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✓ CONFIDENCE FILTERING

Threshold-based validation

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📊 VISUALIZATION

Annotated image rendering

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✅ OUTPUT

Final detection results

3. Architecture Components

3.1 Input Image

Purpose: Serves as the entry point for the detection pipeline, accepting raw visual data from various sources including cameras, file systems, or streaming feeds.

Input: Image files in standard formats (JPEG, PNG, BMP) or video frames with dimensions typically ranging from 640×640 to 1920×1080 pixels.

Output: Raw pixel data matrix passed to the preprocessing stage for normalization and preparation.

Advantages: Supports multiple input sources and formats, enabling flexible deployment across different hardware configurations and use cases.

3.2 Image Preprocessing

Purpose: Transforms raw input images into a standardized format optimized for neural network processing, ensuring consistent model performance across varying input conditions.

Input: Raw image data with variable dimensions, color spaces, and quality levels.

Output: Normalized tensor with fixed dimensions (typically 640×640), scaled pixel values (0-1 range), and standardized color channels (RGB).

Advantages: Improves model accuracy through normalization, reduces computational overhead via resizing, handles lighting variations, and enables batch processing for efficiency.

3.3 YOLOv8 Model

Purpose: Performs real-time object detection using state-of-the-art deep learning architecture, identifying bottle instances and their spatial locations within the image.

Input: Preprocessed image tensor with shape [batch_size, 3, 640, 640] containing normalized RGB values.

Output: Detection tensor containing class predictions, bounding box coordinates (x, y, width, height), and confidence scores for each detected object.

Advantages: Achieves high accuracy with real-time performance (30+ FPS), supports transfer learning for custom datasets, provides end-to-end detection without region proposals, and offers multiple model sizes for different hardware constraints.
